

# COMP SCI 4ZP6A / 4ZP6B

## Computer Science Capstone

Instructor: Mehdi Moradi

*This course outline contains important information that will affect your grade. You should retain and refer to this outline throughout the term. It is your responsibility to be familiar with its contents.*

### 1 Administrative Details

*Note: For security reasons, physical office location information is excluded from this outline.*

#### 1.1 Instructor

- **Instructor:** Mehdi Moradi
- **E-mail:** [moradm4@mcmaster.ca](mailto:moradm4@mcmaster.ca)
- **Office Hours / Support Sessions:** By appointment. The tutorial times (Monday mornings, 9:30 AM – 11:30 AM) are used for team review/support sessions by the instructor (complemented by adding class times when there is higher demand). A sign-up sheet will be posted on Microsoft Teams for booking support/review calls. Teams must sign up at least twice each semester as a team. Until teams are formed, tutorial slots on Monday Sep 14 and Sep 21 will be open drop-in sessions on the General Channel of Teams. Starting Monday Sep 28, 15-minute slots will be booked via the sign-up sheet **by teams**.

#### 1.2 Teaching Assistants (TAs)

Your team will be assigned a TA to support you. TAs will post their own office hours and meeting schedules.

- **TA 1:** Amir Hossein Sabour ([saboua4@mcmaster.ca](mailto:saboua4@mcmaster.ca))
- **TA 2:** Sahib Khokhar ([khokhs5@mcmaster.ca](mailto:khokhs5@mcmaster.ca))
- **TA 3:** Tom Overall ([overallt@mcmaster.ca](mailto:overallt@mcmaster.ca))

#### 1.3 Session and Term

- **Academic Year / Session:** 2026–2027

- **Term:** Fall 2026 and Winter 2027 (Full Year Course: 4ZP6A and 4ZP6B)

## 2 Calendar Description

## 3 Mission

**Formal Version:** Students, in teams, undertake a substantial project in an area of computer science by performing each step of the software life cycle.

**Informal Version:** To give you an opportunity to experience “building” a system or “solving a real-world problem” while also managing the messy realities of teamwork in an imperfect world. That is why we insist on teams of at least 5 students in size. The final grading of the work will be based on a working solution. Your TAs will need to be able to see a working system (application, website, plugin, ... whatever you build should be deployed and usable for a third party).

### 3.1 Intended Learning Outcomes

- Knowledge of product planning process, including Software Requirement Specification (SRS), Project Plan and Design Documents.
- Learn to work as part of a software development team.
- Learn and implement a prioritization system for the features of your proposed project/product.
- Deliver on the requirements as defined in your Requirement Specification.

### 3.2 Graduate Attributes and Indicators

This course is NOT part of an Engineering program.

- **Graduate Attribute:** Knowledge

## 4 Materials & Fees

### 4.1 Required Resources

- **Textbook:** NO TEXTBOOK IS REQUIRED FOR THIS COURSE (\$0.00).
- **Departmental Financial Support & Reimbursements:**
  - CAS will reimburse groups for approved expenses, up to a maximum of \$125/group. All expenses submitted for reimbursement must be approved by your TA. Reimbursement requests must be sent to [casadmin@mcmaster.ca](mailto:casadmin@mcmaster.ca) and must include proof of the TA's written approval, all necessary receipts (including HST number), and a CC to the instructor.
  - CAS will cover the cost of printing EXPO posters up to \$60/group (details to come later).

## 5 Optional Resources

None.

## 6 Engineering Tools and Lab Experience

- **Departmental Server & Computing Infrastructure:** The Department of Computing and Software maintains its own server and end-user computing infrastructure. Students have access to a virtual desktop cluster from terminals in labs as well as their own devices. On the server side, a GPU cluster is available for LLM computing, as well as a private cloud for groups that require a virtual server for their project.
- **App Store Accounts:** CAS has set up organizational accounts with both the Apple App Store and Google Play for use within your Capstone course. (For more details, visit the CAS tech support webpage – accessible on campus or via VPN).
- **GitHub & Version Control:**
  - Each team needs to create and maintain a GitHub repository.
  - Use your GitHub repo for code tracking and version control. Commit code under the folder src. The instructor and TAs will monitor code commits. If there is a dispute about lack of contributions by a team member, code commits and meeting attendance will be investigated to verify contribution levels.
  - Maintain a copy of each deliverable submission in your repository under a folder named docs. This version will become part of the final evaluation package and can be revised during the year.
  - In your repository, create a folder for meetings. Each time you meet as a team, create a brief text document in this folder listing attendee names and the meeting agenda.
- **Learning Management & Communication Platforms:**
  - **Avenue to Learn:** All deliverables (except source code) must be submitted on Avenue by 11:59 PM EST on the deadline day. Only PDF files are accepted.
  - **Microsoft Teams:** Considered an official channel of communication. Students are expected to routinely check announcements.

## 7 Course Grading

### 7.1 Team & Project Guidelines

- **Team Size:** 5–7 students. These upper and lower limits will be strictly enforced.
- **Team Policy:** The faculty does not allow 4ZP6 students to join teams from other engineering capstones. You must be part of a team in this course. No exceptions.

- **Selecting a Project and Potential Supervisor:**
  - Students can select a project from the list of project proposals shared between SE and CS courses (posted under Files in the Teams General channel before classes start), or propose their own project.
  - Student-proposed projects may stem from interactions with CAS faculty, outside-CAS faculty, industry, or original ideas (original concept, continuation of an older topic, or repeating an older project with significant improvements).
  - Projects can be carried out and supervised off-campus (e.g., resulting from summer employment). Discuss any industry collaborations with the instructor and the McMaster Industry Liaison Office (MILO) as soon as possible.
  - If selecting an externally or faculty-proposed project, the proposer can supervise technical aspects, but grading remains the responsibility of the course instructor.
  - **Academic vs Conceptually Simple Projects:** In complex academic research projects, evaluation emphasizes finding a solution while maintaining documentation standards. For conceptually simpler solutions, students are expected to demonstrate excellence in collecting requirements, human-computer interaction, rigorous testing, and supervisor communication.

## 7.2 Timeline of Deliverables & Grade Breakdown

| Milestone / Deliverable                                                                                                                                                                                              | Deadline     | % of Overall Grade |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------|
| <b>Team and project selection:</b> <ul style="list-style-type: none"> <li>• Form a team</li> <li>• Select a project (title and short description)</li> <li>• Create GitHub repo and add TA and instructor</li> </ul> | Sep 21, 2026 | 5%                 |
| Potential revision of project descriptions/selections based on feedback for student-proposed projects (project topics frozen after this date)                                                                        | Sep 28, 2026 | 0%                 |
| <b>Unified Software Requirement Specification (SRS) and Project Plan documentation</b>                                                                                                                               | Oct 9, 2026  | 15%                |
| <b>Proof of concept demonstration:</b> Two-minute video clip.<br><i>(Note: Tutorial and class time combined for a 3-hour</i>                                                                                         | Nov 23, 2026 | 10%                |

| Milestone / Deliverable                                                                                                                                    | Deadline      | % of Overall Grade |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|
| <i>session; location TBD for tutorial time)</i>                                                                                                            |               |                    |
| <b>Design and V&amp;V document</b>                                                                                                                         | Jan 25, 2027  | 10%                |
| <b>Final demonstration:</b> Two-minute video clip.<br><i>(Note: Tutorial and class time combined for a 3-hour session; location TBD for tutorial time)</i> | April 5, 2027 | 5%                 |
| <b>Final submission:</b> Source code repository + finalized design document + User guide (all on GitHub) + working app/website + Poster PDF                | April 5, 2027 | 50%                |
| <b>EXPO Demonstration and poster</b>                                                                                                                       | TBD           | 5%                 |
| <b>Total</b>                                                                                                                                               | -             | <b>100%</b>        |

### 7.3 Evaluation Criteria for Final Submission (50%) & High Grade Guide

Document submissions are graded against instructions. The final 50% submission (and overall guidance throughout) is governed by:

- **Complexity (40%):** A project representing substantive effort beyond simple scripts (benchmarked at ~6 engineers working 10 hours/week over 6 months):
  - *Low (0–20%):* Think static website; even if polished, falls on the lower end.
  - *Medium (20–40%):* Well-executed app or service. Innovative tech is not required, but software engineering and execution must be top-tier.
  - *High (30–40%):* Well-built project incorporating complex algorithms.
- **Completeness (40%):**
  - Hitting all P0 and P1 targets.
  - System must actually work so TAs can run and test it independently.
  - If building a tool that uses ML/AI, a legitimate, sound metric must be implemented to track performance.
- **Professionalism (20%):** Soft skills, project logistics, and communication:
  - Meet with the supervisor twice per semester and TAs at least once (students must sign up). Failure to book team meetings will result in lost points.
  - Adhere to deadlines and guidelines, demonstrate openness to feedback, keep supervisors informed, and attend mandatory demo/poster sessions.

## 8 Course Schedule

### 8.1 Weekly Structure

- **Class Time:** Mon, Wed, Thu 1:30 PM – 2:20 PM (*Used for **in-person** lectures only when advertised below as mandatory; otherwise utilized as dedicated team working time*).
- **Tutorial Time:** Monday 9:30 AM – 11:30 AM (*Used for meeting the instructor; sign up **as a team** at least twice a semester on the General channel sign-up sheet. Meetings will happen on the General Channel of Teams*).

### 8.2 Key Dates & In-Person Attendance Sessions

- **Mandatory In-Person Lectures (1:30 PM – 2:20 PM):**
  - Week 1: Wednesday Sep 9 and Thursday Sep 10, 2026
  - Thursday Sep 24, 2026
- **Mandatory In-Person Presentation Events:**
  - **Proof of Concept Presentation:** Monday Nov 23, 2026 (3 hours total: 9:30 AM – 11:20 AM Location TBD; 1:30 PM – 2:20 PM in regular classroom).
  - **Final Course Presentations:** Monday April 5, 2027 (3 hours total: 9:30 AM – 11:20 AM Location TBD; 1:30 PM – 2:20 PM in regular classroom).

## 9 Policy on Missed Work, Extensions and Late Penalties

*Refer to Section 10.7 for university policy on relief for missed academic term work.*

## 10 Approved Advisory Statements

### 10.1 Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the

university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at

<https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/>.

The following illustrates only three forms of academic dishonesty:

- Plagiarism, e.g., the submission of work that is not one's own or for which other credit has been obtained.
- Improper collaboration in group work, including not doing your share of the work.
- Copying or using unauthorized aids in tests and examinations.

## **10.2 Authenticity/Plagiarism Detection**

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to [www.mcmaster.ca/academicintegrity](http://www.mcmaster.ca/academicintegrity).

## **10.3 Course with an On-Line Element**

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, usernames for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

## **10.4 On-Line Proctoring**

Some courses may use on-line proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests

or exams. This software may be required to be installed before the test/exam begins.

## **10.5 Conduct Expectations**

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online. It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on on-line platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

## **10.6 Academic Accommodation for Students with Disabilities**

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or [sas@mcmaster.ca](mailto:sas@mcmaster.ca) to make arrangements with a Program Coordinator. For further information, consult McMaster University’s Academic Accommodation of Students with Disabilities policy.

## **10.7 Requests for Relief for Missed Academic Term Work**

In the event of an absence for medical or other reasons, students should review and follow the McMaster Student Absence Form Policy (MSAF Policy).

## **10.8 Academic Accommodation for Religious, Indigenous or Spiritual Observances (RISO)**

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar’s Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

## **10.9 Copyright and Recording**

Students are advised that lectures, demonstrations, performances, and any other course

material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors. The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

## **10.10 Extreme Circumstances**

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.

# **11 Additional Statements**

This section summarizes additional policy statements for the course.

## **11.1 Generative AI**

- Students may use generative AI in this course in accordance with the guidelines outlined for each assessment, and so long as the use of generative AI is referenced and cited. Use of generative AI outside assessment guidelines or without citation will constitute academic dishonesty. It is the student's responsibility to be clear on the limitations for use for each assessment and to be clear on the expectations for citation and reference and to do so appropriately.
- Guideline on how to describe use of Generative AI: The use has to be documented with a brief description of the nature of the use (example: "I used Claude to brainstorm ideas for X, and used Gemini to edit the text").
  - This is NOT an excuse for bad work! AI can help. But you are responsible for your work, its quality and accuracy, and making sure AI does not infringe on copyrighted material or plagiarize. You alone are responsible for your work.

## **11.2 Improving the Course**

Ideas to improve the course are always welcome. Moreover, if you have problems in the course, please contact the instructors as early as possible.

## **11.3 Discrimination**

The Faculty of Engineering is committed to creating an environment in which students of all

genders, cultures, ethnicities, races, sexual orientations, abilities, and socioeconomic backgrounds have equal access to education and are welcomed and treated fairly. If you have any concerns regarding inclusion in our Faculty, in particular if you or one of your peers is experiencing harassment or discrimination, you are encouraged to contact the Department Chair, Associate Chair Undergraduate, Academic Advisor, or the Equity and Inclusion Office.